Spectral Predictors for Tseitin Hardness: A Physics-Motivated Exploration

Preliminary Conjectures and Empirical Observations

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EXPLORATORY PREPRINT — NOT PEER-REVIEWED

This work presents **conjectures and preliminary observations** on spectral predictors for proof complexity. No new theorems are proved. Empirical validation is limited to small illustrative instances ($n \leq 80$). This document invites discussion, testing, and critique.

Abstract

We explore a physics-first viewpoint on propositional proof complexity, motivated by Landauer's principle [1]: information processing has an inherent physical cost. For Tseitin contradictions over quasi-regular graphs [3, 6], we argue that any solver must process a minimal amount of *structural information* depending jointly on local density and global connectivity.

We introduce the predictor

$$\widehat{\deg}(G) := \frac{n}{\sqrt{1/d + 1/\lambda_2}},$$

where n is the number of vertices, d the average degree, and λ_2 the second-smallest eigenvalue of the normalized Laplacian [2]. We formulate corresponding Kolmogorov-type information conjectures [7], conjectural decision-tree lower bounds, and connections to Polynomial Calculus degree.

Empirically, we observe that small structural perturbations can induce sharp, solver-dependent transitions in SAT solver behavior at fixed (n, d), notably under Watts–Strogatz rewiring. We interpret these transitions as evidence for underlying *information thresholds* governing solver difficulty.

Note. This is an exploratory preprint presenting conjectures and preliminary empirical observations. No new theorems are proved.

Contributions. This exploratory work offers:

- A physics-motivated *conjectural* framework linking spectral structure to proof complexity for Tseitin formulas;
- A structural predictor $\widehat{\deg}(G)$ combining local density and global connectivity;
- Empirical evidence of a *structural phase transition* in SAT solver behavior under small graph perturbations at fixed (n, d);
- A preliminary connection between these transitions and information-theoretic interpretations of solver complexity;

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- A roadmap of conjectures linking spectral graph theory, decision-tree complexity, Polynomial Calculus, and Kolmogorov complexity.

No new theorems are proved.

1 Introduction

The cost of distinguishing SAT from UNSAT is fundamentally informational. A solver must extract, manipulate, and retain a minimal volume of structural information about the instance. Following Landauer's principle [1], we take seriously the idea that this informational cost has a structural footprint in the input.

Tseitin contradictions [3, 6] are among the canonical hard families in propositional proof complexity. They encode parity constraints on a graph G and are unsatisfiable exactly when the total charge is odd. On expander graphs, Tseitin formulas yield strong lower bounds in Resolution [5] and algebraic systems such as Polynomial Calculus (PC) [4].

Meanwhile, spectral graph theory [2, 8] provides a robust way to quantify connectivity via the spectrum of the normalized Laplacian, with the spectral gap λ_2 tightly related to edge expansion and mixing.

In this preprint we propose a structural predictor,

$$\widehat{\deg}(G) = \frac{n}{\sqrt{1/d + 1/\lambda_2}},$$

combining local density (d) and global connectivity (λ_2) in a harmonic fashion. We conjecture that $\widehat{\deg}(G)$ approximates the minimal information budget that any reasoning process must handle to refute $\mathrm{Tseitin}(G, \chi)$. We phrase this in time-bounded Kolmogorov terms [7] and as conjectured lower bounds in decision-tree and algebraic proof models.

Our goal here is not to present a complete lower-bound proof, but to propose a coherent structural scale that can be probed both analytically and empirically, and which appears to interpolate naturally between known behaviors on cycles, grids, expanders and dense graphs.

Even when the degree is fixed, the global geometry of the constraint graph can vary significantly depending on the generation model, as illustrated in Figure 1.

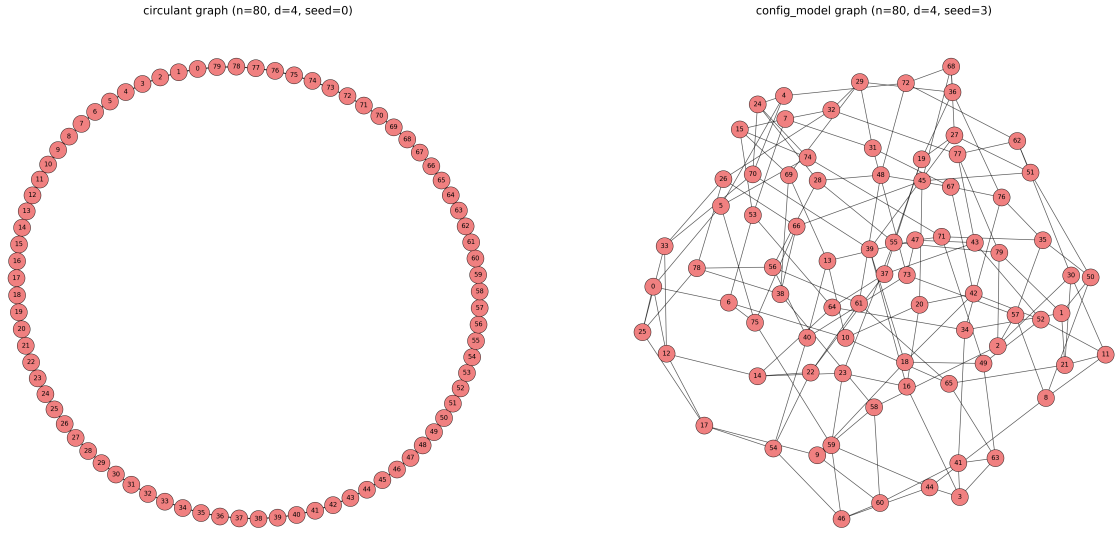

Figure 1: Degree-regular graphs with identical parameters ($n = 80$, $d = 4$) generated using two different models. **Left:** circulant graph (highly symmetric, homogeneous geometry). **Right:** configuration-model graph (irregular geometry with local bottlenecks). Despite identical degree sequences, the resulting structures differ substantially at the geometric level.

While the spectral predictor $\widehat{\deg}(G)$ provides a guiding theoretical hypothesis, our experimental results primarily highlight strong structural sensitivity phenomena. In particular, we observe that very small perturbations in graph structure can induce abrupt transitions in solver behavior, suggesting the existence of underlying information thresholds.

2 Graphs, spectrum, and conventions

Let $G = (V, E)$ be a simple graph with $|V| = n$ and average degree d. The normalized Laplacian is

$$\mathcal{L} = I - D^{-1/2} A D^{-1/2},$$

with eigenvalues

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n \leq 2,$$

where λ_2 is the second-smallest eigenvalue (the spectral gap) [2].

Remark (Cheeger inequality). Let $\phi(G)$ denote edge conductance. Cheeger-type inequalities for the normalized Laplacian [2] give

$$\frac{\phi(G)^2}{2} \leq \lambda_2 \leq 2\phi(G),$$

supporting the use of λ_2 as a proxy for expansion.

We focus on quasi-regular families (bounded ratio between maximum and minimum degrees), and we view G as the constraint graph of a Tseitin formula [3, 6].

3 MPCC and Kolmogorov formulation

We now phrase the Minimal Proof Compression Conjecture (MPCC) and a Kolmogorov-based variant.

Conjecture 1 (MPCC)**.** *For any quasi-regular family of graphs, any solver deciding* $\mathrm{Tseitin}(G, \chi)$ *must process at least*

$$c \cdot \frac{n}{\sqrt{1/d + 1/\lambda_2}}$$

bits of structural information about G*, for some constant* $c > 0$*, robust under preprocessing and natural re-encodings.*

Kolmogorov formulation. Following Li–Vitányi [7], let K^t denote time-bounded Kolmogorov complexity with side information. For a canonical encoding $\mathrm{enc}(G)$, set

$$K^t(G \mid (n, d, \lambda_2)) = \min\{|p| : U(p, (n, d, \lambda_2)) = \mathrm{enc}(G) \text{ in time } \leq t(|G|)\}.$$

Conjecture 2 (Kolmogorov–MPCC)**.** *There exist* $c > 0$ *and a polynomial* $t(\cdot)$ *such that for all quasi-regular graphs* G*,*

$$K^t(G \mid (n, d, \lambda_2)) \;\geq\; c \cdot \frac{n}{\sqrt{1/d + 1/\lambda_2}} - o(n).$$

4 Decision-tree lower bounds

We informally sketch a decision-tree model for refuting Tseitin contradictions and formulate our lower-bound claims as conjectures, since a fully detailed proof is not included in this preprint.

Let $\mathrm{R}_0^{\mathrm{DT}}(G)$ denote the zero-error randomized decision-tree complexity of refuting $\mathrm{Tseitin}(G, \chi)$, where each query requests local information about edges and incident constraints.

Conjecture 3 (DT lower bound for $d = O(1)$)**.** *For any bounded-degree quasi-regular graph* G*, there exists* $c > 0$ *such that*

$$\mathrm{R}_0^{\mathrm{DT}}(G) \;\geq\; c\, n \sqrt{\lambda_2}.$$

Conjecture 4 (Harmonic-bound template)**.** *For any quasi-regular graph* G*, there exists* $c > 0$ *such that*

$$\mathrm{R}_0^{\mathrm{DT}}(G) \;\geq\; c \cdot \frac{n}{\sqrt{1/d + 1/\lambda_2}}.$$

These conjectures are motivated by Yao-style arguments and the behavior of expander graphs [8].

5 Polynomial Calculus connections

Polynomial Calculus (PC) is an algebraic proof system where propositional formulas are translated into polynomial equations over a field and refutations proceed by algebraic derivations [4, 6]. For Tseitin contradictions over expanders and fields of characteristic not equal to 2, Razborov [4] proved linear lower bounds on PC degree.

These results fit the qualitative picture suggested by $\widehat{\deg}(G)$: stronger expansion implies higher degree, and thus higher algebraic complexity.

Conjecture 5 (PC-degree $\Rightarrow$ touch complexity)**.** *If* $\deg_{\mathsf{PC}}(\mathrm{Tseitin}(G)) \geq r$ *then*

$$\mathrm{Touch}_{\mathsf{PC}}(G) \;\geq\; cr$$

for some absolute constant $c > 0$*, where* $\mathrm{Touch}_{\mathsf{PC}}(G)$ *is a suitable measure of how many constraint occurrences any PC derivation must "touch".*

We expect $\mathrm{Touch}_{\mathsf{PC}}(G)$ to scale with $\widehat{\deg}(G)$, linking PC-degree lower bounds to the MPCC perspective.

6 Calibrating the predictor $\widehat{\deg}(G)$

We define

$$\widehat{\deg}(G) := \frac{n}{\sqrt{1/d + 1/\lambda_2}}.$$

This form matches natural scales on standard graph families [2, 8]:

Family	d	λ_2	$\widehat{\deg}(G)$	Comment
Cycle C_n	2	$\Theta(1/n^2)$	$\Theta(1)$	1-D weak
Grid $m \times m$	$O(1)$	$\Theta(1/n)$	$\Theta(\sqrt{n})$	2-D medium
Expander	$O(1)$	$\Theta(1)$	$\Theta(n)$	strong
Complete K_n	n	1	$\Theta(n)$	dense

7 Empirical Results

We test the alignment between $\widehat{\deg}(G)$ and practical SAT difficulty for Tseitin CNFs. Families include cycles C_n, grids, complete bipartite graphs, and random regular graphs. CNFs are generated by constructing G and encoding XOR constraints at each vertex into 3-CNF using auxiliary variables.

We present controlled experiments investigating the impact of graph structure on solver behavior, as well as preliminary evidence relating this behavior to the spectral predictor $\widehat{\deg}(G)$.

7.1 Experimental pipeline and structure dependence

We use a reproducible C++ pipeline in which each instance is fully determined by $(n, d, \texttt{seed}, \texttt{graph_mode}, p)$. The pipeline generates the graph, builds the corresponding Tseitin CNF, computes a byte-level hash of the DIMACS file (`cnf_hash`), and runs a SAT solver under fixed resource limits.

We evaluate two CDCL solvers:

- **Kissat** (modern high-performance solver),
- **MiniSat** (baseline reference solver).

All runs use a 60 s wall-clock timeout. Each configuration is evaluated over 5 independent seeds.

At fixed size $n = 80$ and degree $d = 4$, we observe strong structure dependence even when (n, d) are held constant. In particular, graph generation models with identical degree sequences can yield dramatically different solver behavior.

7.2 Watts–Strogatz transition

We study d-regular graphs generated using the Watts–Strogatz rewiring model, with rewiring probability

$$p \in \{0, 0.01, 0.02, 0.05\}.$$

All instances correspond to Tseitin contradictions and are UNSAT by construction.

Observed transition. We observe a sharp transition in solver behavior, which we refer to as a *structural phase transition.*

For Kissat:

- $p = 0$: easy regime (median 22 ms)

- $p = 0.01$: heavy-tailed regime (median 6305 ms)
- $p \geq 0.02$: all runs timeout

For MiniSat:

- $p = 0$: easy regime (median 48 ms)
- $p \geq 0.01$: all runs timeout

The abruptness of the transition, occurring between $p = 0.01$ and $p = 0.02$, suggests a threshold phenomenon rather than a smooth degradation.

Interpretation. These results indicate that small structural perturbations can dramatically alter the effective search landscape of CDCL solvers.

The shift in transition threshold between MiniSat and Kissat suggests that this phenomenon depends not only on instance structure, but also on the interaction between graph geometry and solver heuristics.

In particular, the heavy-tailed regime observed for Kissat at $p = 0.01$ suggests the presence of an intermediate phase in which instances remain solvable, but require highly variable amounts of computational effort.

Relation to the spectral predictor. Although n and d are fixed, rewiring significantly alters the global structure of the graph, likely affecting expansion and the spectral gap λ_2.

While these results are consistent with the hypothesis that $\widehat{\deg}(G)$ captures structural hardness, we do not yet compute λ_2 for these instances. Establishing a direct quantitative link between $\widehat{\deg}(G)$ and the observed transition remains an important direction for future work.

7.3 Runtime distributions

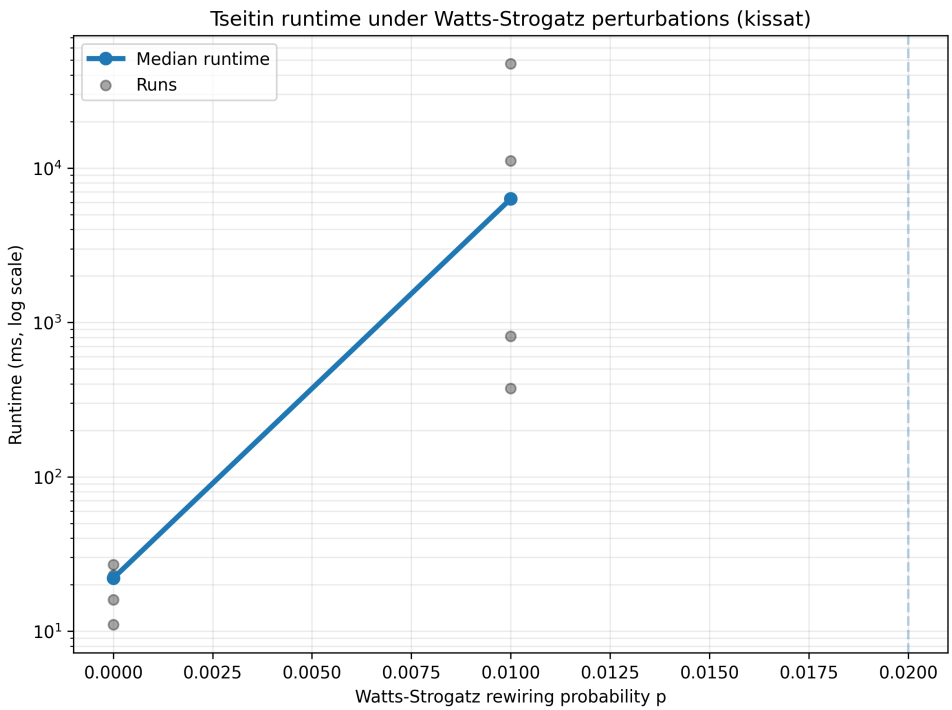

Figure 2: Kissat runtimes under Watts–Strogatz perturbations. Each point corresponds to one seed; the blue curve shows the median. A sharp increase appears at $p = 0.01$, followed by systematic timeout.

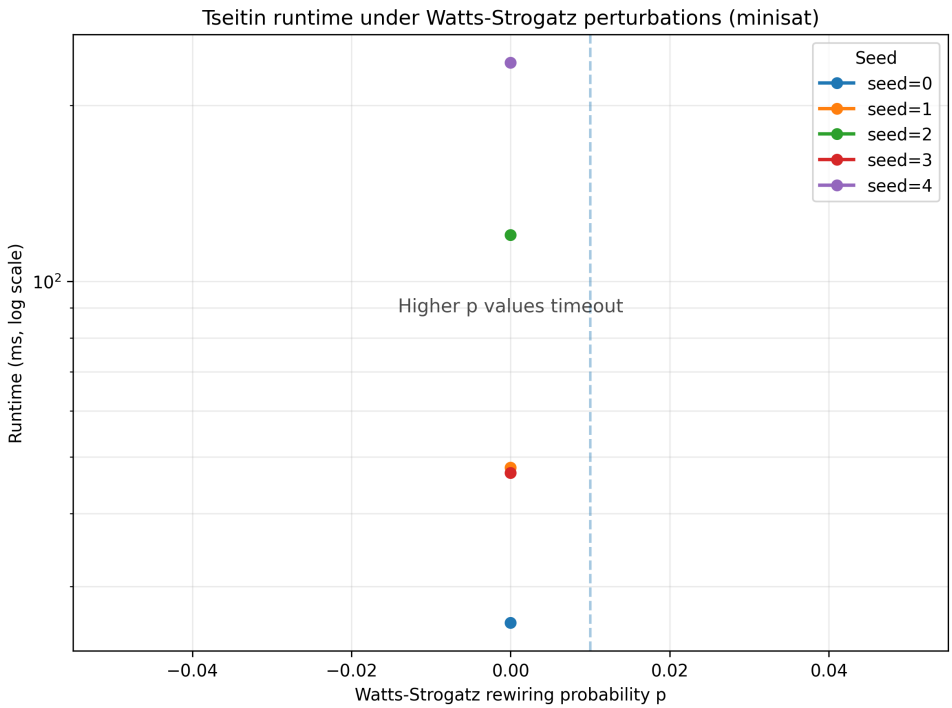

Figure 3: MiniSat runtimes for the same experiment. The transition occurs earlier, with immediate timeout at $p = 0.01$.

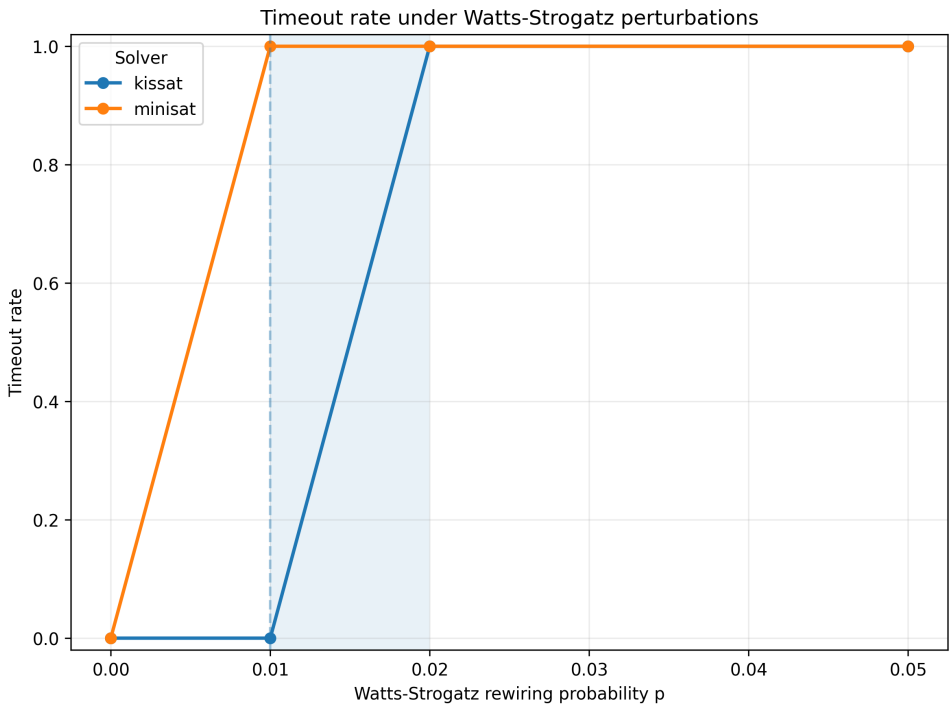

Figure 4: Timeout rate as a function of rewiring probability p. MiniSat reaches full timeout at $p = 0.01$, whereas Kissat does so at $p = 0.02$.

7.4 Spectral predictor vs runtime

We compare $\widehat{\deg}(G)$ with observed solver runtimes on small benchmark families.

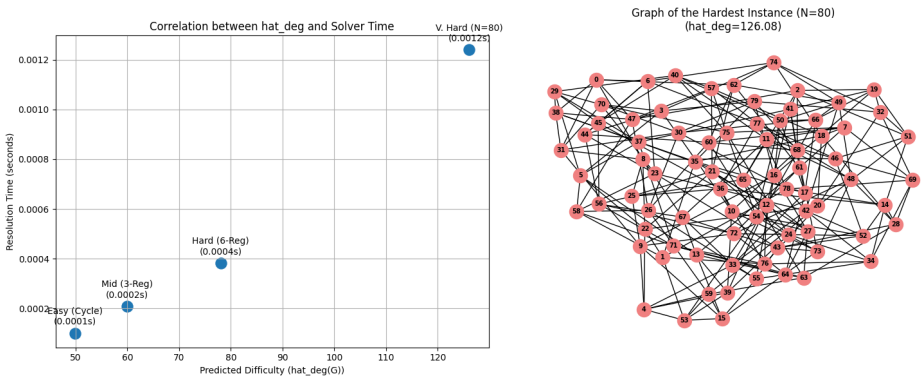

Figure 5: Correlation between $\widehat{\deg}(G)$ and solver runtime. Higher $\widehat{\deg}(G)$ values tend to correspond to harder instances.

These results suggest that $\widehat{\deg}(G)$ captures a meaningful structural component of practical SAT difficulty, although further large-scale validation is required.

8 Discussion: Structural Phase Transition and Information Thresholds

The observed transition under Watts–Strogatz perturbations suggests the existence of structural thresholds governing solver behavior.

In highly regular graphs ($p = 0$), the structure appears compressible and efficiently exploitable by CDCL heuristics. Introducing a small amount of randomness disrupts this structure, leading to a sharp increase in computational effort.

This behavior is consistent with the intuition underlying the MPCC conjecture: solvers must process a minimal amount of structural information before deriving a contradiction.

The existence of solver-dependent thresholds further suggests that this phenomenon arises from the interaction between instance structure and solver heuristics, rather than from a purely combinatorial property of the instance.

While these interpretations remain speculative, the reproducibility and sharpness of the observed transition provide strong motivation for further investigation.

9 Limitations and Future Directions

Theoretical gaps. All main claims (Conjectures 1–5) remain unproved. Establishing even a partial result—for example, PC-degree(Tseitin(G)) $\geq \Omega(\widehat{\deg}(G))$ for a restricted graph family—would significantly strengthen the framework. Clarifying the relation between $\widehat{\deg}(G)$ and known measures such as Resolution width [5] or treewidth is an important open direction.

Experimental limitations. The empirical validation in this preprint is limited in scope:

- small instance sizes ($n \leq 80$).
- few data points per family (on the order of 4–10 instances).
- no formal statistical analysis (regression, confidence intervals).
- no comparison with alternative predictors (e.g. treewidth, Resolution width, expansion alone, clause-learning statistics).
- apparent outliers (e.g. 6-regular $n = 80$) remain unexplained.
- no claim of average-case hardness or asymptotic lower bounds is made.
- the observed transition is reported at a single (n, d) and with a fixed timeout, mapping its dependence on n, d, and solver settings remains future work.
- In particular, we do not yet compute the spectral gap λ_2 for the Watts–Strogatz instances, and therefore cannot directly evaluate $\widehat{\deg}(G)$ in this regime.
- The observed structural phase transition is documented at a single (n, d) scale, and its dependence on graph size, degree, and solver configuration remains to be studied.

A convincing validation would require: (i) at least 100 instances per family, scaling to $n \geq 10^3$; (ii) multiple random seeds per configuration; (iii) formal regression analysis (e.g. log-time explained by $\widehat{\deg}(G)$); and (iv) head-to-head comparison with known complexity measures. In particular, we do not yet compute the spectral gap λ_2 for the Watts–Strogatz instances, and therefore cannot directly evaluate $\widehat{\deg}(G)$ in this regime.

Open questions. We highlight a few questions that we believe are both natural and approachable:

- Does $\widehat{\deg}(G)$ provably lower-bound PC-degree or Resolution width for some nontrivial graph families?
- How does $\widehat{\deg}(G)$ compare empirically to treewidth or pathwidth as a hardness predictor across standard SAT benchmarks?

- Can the Kolmogorov formulation (Conjecture 2) be made rigorous via explicit incompressibility arguments for random or pseudorandom graph families?

We view this work as a *hypothesis-generating* exercise and encourage the community to test, extend, or refute these ideas.

10 Licensing

Experimental code (Python scripts) is released under the **MIT License**. This document and embedded figures are under the **CC BY 4.0** license.

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